

Xianrui Luo

Homepage

Google Scholar

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BIOGRAPHY

I am a postdoctoral researcher (Shuimu Scholar) at the Institute of Future Cities and Infrastructures, Tsinghua University, working with [Prof. Dongping Fang](#). I received my Ph.D. degree from the School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, supervised by [Prof. Zhiguo Cao](#). My research interests lie in (1) 3D vision and image manipulation, with a particular emphasis on computational photography; (2) multimodal language models and their applications in construction automation.

EMPLOYMENT

Postdoctoral Researcher Institute of Future Cities and Infrastructures, Tsinghua University Advisor: Prof. Dongping Fang . I lead a project on multi-modal LLMs in construction safety.	Aug 2025 – Now
Project Officer S-Lab for Advanced Intelligence, Nanyang Technological University Advisor: Prof. Guosheng Lin . I lead a project on high-fidelity human avatar modeling.	Nov 2023 – Nov 2024

EDUCATION

Huazhong University of Science and Technology, China Doctor of Artificial Intelligence Thesis: "All-in-Focus Novel View Synthesis from Defocus-Blurred Images"	Sep 2020 – Jun 2025 GPA: 90.19
Huazhong University of Science and Technology, China Bachelor of Automation	Sep 2016 – Jun 2020 GPA: 89.2

PUBLICATIONS

Journal Articles

Toward Fine-Grained Construction Safety Inspection via Vision-Language Grounded Reasoning Xianrui Luo , Chengqian Li, Botao Gu, Dongping Fang	Advanced Engineering Informatics 2026 [Paper]
Dual-Camera All-in-Focus Neural Radiance Fields Xianrui Luo , Zijin Wu, Juewen Peng, Huiqiang Sun, Zhiguo Cao, Guosheng Lin	IEEE TPAMI 2025 [Paper] [Code]
Defocus to Focus: Photo-realistic Bokeh Rendering by Fusing Defocus and Radiance Priors Xianrui Luo* , Juewen Peng*, Ke Xian, Zijin Wu, Zhiguo Cao (*Equal Contribution)	Information Fusion 2023 [Paper] [Code]
Point-and-Shoot All-in-Focus Photo Synthesis From Smartphone Camera Pair Xianrui Luo , Juewen Peng, Weiyue Zhao, Ke Xian, Hao Lu, Zhiguo Cao	IEEE TCSVT 2023 [Paper]
BokehMe++: Harmonious Fusion of Classical and Neural Rendering for Versatile Bokeh Creation Juewen Peng, Zhiguo Cao, Xianrui Luo , Ke Xian, Wenfeng Tang, Jianming Zhang, Guosheng Lin	IEEE TPAMI 2024 [Paper]

Conference Papers

Dynamic Neural Radiance Field From Defocused Monocular Video Xianrui Luo , Huiqiang Sun, Juewen Peng, Zhiguo Cao	ECCV 2024 [Paper] [Code]
Bokeh Rendering from Defocus Estimation Xianrui Luo* , Juewen Peng*, Ke Xian, Zijin Wu, Zhiguo Cao (*Equal Contribution)	ECCVW 2020 [Paper]
BokehFlow: Depth-Free Controllable Bokeh Rendering via Flow Matching Yachuan Huang, Xianrui Luo , Qiwen Wang, Liao Shen, Jiaqi Li, Huiqiang Sun, Zihao Huang, Wei Jiang, Zhiguo Cao	AAAI 2026 [Paper]
BokehCrafter: Taming Video Diffusion Models for Controllable Bokeh Rendering Qiwen Wang, Liao Shen, Jiaqi Li, Tianqi Liu, Huiqiang Sun, Zihao Huang, Yachuan Huang, Xianrui Luo , Zhiguo Cao	AAAI 2026 [Paper]
Fast Full-frame Video Stabilization with Iterative Optimization Weiyue Zhao, Xin Li, Zhan Peng, Xianrui Luo , Xinyi Ye, Hao Lu, Zhiguo Cao	ICCV 2023 [Paper] [Code]

Selective Bokeh Effect Transformation Juewen Peng, Zhiyu Pan, Chengxin Liu, Xianrui Luo , Huiqiang Sun, Liao Shen, Ke Xian, Zhiguo Cao	CVPRW 2023 [Paper] [Code]
MPIB: An MPI-Based Bokeh Rendering Framework for Realistic Partial Occlusion Effects Juewen Peng, Jianming Zhang, Xianrui Luo , Hao Lu, Ke Xian, Zhiguo Cao	ECCV 2022 [Paper] [Code]
BokehMe: When Neural Rendering Meets Classical Rendering Juewen Peng, Zhiguo Cao, Xianrui Luo , Hao Lu, Ke Xian, Jianming Zhang	CVPR 2022 [Paper] [Code]
Interactive Portrait Bokeh Rendering System Juewen Peng, Xianrui Luo , Ke Xian, Zhiguo Cao	ICIP 2021 [Paper]

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PROJECT

Shenzhen Construction Engineering Group Funding Project on Safety Inspection - Develop a reinforcement learning-based fine-tuning framework for pretrained vision-language models to improve automated construction safety hazard inspection under image-level weak supervision. - Reformulate safety inspection from direct hazard classification to evidence-based visual reasoning, enabling the model to localize hazard-relevant visual evidence before generating final safety judgments. - Curate and quality-control real-world construction safety inspection data, including label cleaning, duplicate filtering, and benchmark construction for evaluating domain-adapted VLMs.	Sep 2025 – Now
S-Lab Funding Project on Human Avatar Modeling - Build and clean large-scale motion datasets for character animation, improving data quality and reliability for downstream action generation model training. - Develop data capture and synthesis pipelines for high-fidelity human avatar modeling, supporting research on 3D human reconstruction and photorealistic avatar generation.	Nov 2023 – Nov 2024
NTIRE 2023 Challenge on Bokeh Effect Transformation (CVPR2023 Workshop) Winner Award	Jun 2023
Vivo Funding Project on Simulation of Bokeh Effect from DSLR I serve as a core member responsible for realistic bokeh synthesis for smartphone photography, I analyze experimental results and provide ideas to improve visualizations.	Jan 2023 – Dec 2023
AIM 2020 Challenge on Rendering Realistic Bokeh (ECCV2020 Workshop) Runner Up Award	Aug 2020

AWARDS AND HONORS

National Scholarship (1% of all students)	<i>Huazhong University of Science and Technology</i>	2024
Academic Scholarship	<i>Huazhong University of Science and Technology</i>	2020 – 2024
Outstanding Graduates	<i>Huazhong University of Science and Technology</i>	2020
Merit Student (5% of all students)	<i>Huazhong University of Science and Technology</i>	2016, 2023
Best Poster Award	<i>Artificial Intelligence Conference and Entrepreneurs Summit Forum of China's Optics Valley</i>	2022

SKILLS

Programming Language <i>Python, MATLAB, C</i>	Machine Learning Tools <i>PyTorch, OpenCV</i>	English <i>TOEFL 109, GRE 325 (AW 4.0)</i>
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